What is an API

API stands for Application Programming Interface.

It is a set of rules that allows two software applications to communicate with each other.

Think of an API as a **messenger** that takes a request from one application, tells another application what to do, and then brings the response back.

In simple words:

API is a messenger that carries requests from one application to another and brings back the response.

Why API is needed?

- To connect different applications
- To share data safely
- To reuse features (Login with Google, online payments, etc.)
- To make mobile apps work with servers
- To reduce development time

Example:

- You (customer) → send order
- Waiter (API) → carries the order
- Kitchen (server) → prepares food
- Waiter returns the food to you

That means:

Defines *how* different software should talk

Uses **endpoints (URLs)**

Sends/receives **data** (mostly JSON)

Uses **methods** like GET, POST, PUT, DELETE

Works between **client** (frontend) and **server** (backend)

Common Examples of APIs

- Payment APIs
  - Razorpay, PayPal — allow apps to accept online payments.
- Login APIs
  - Google, Facebook — allow "Login with Google/Facebook".
- Maps API
  - Google Maps API shows locations on apps.
- Weather API

Types of APIs

- **Web APIs** (REST, SOAP, GraphQL)
- **Library APIs** (Django API, Android API)
- **Operating System APIs** (Windows API, Android API)
- **Hardware APIs** (Camera, Bluetooth APIs)

Django REST framework

DRF (Django REST Framework) is a powerful library built on top of Django that helps you create **APIs** easily.

While normal Django is used to create **webpages (HTML)**,
DRF is used to create APIs (mostly JSON) so that:

- Mobile apps
- Frontend frameworks (React, Angular, Vue)
- Other software systems

DRF converts your Django models into APIs that return data like JSON, not HTML.

➢ Features of DRF

1. Serializers

   Convert Django model data → JSON and JSON → Python objects.
2. Views & ViewSets
3. Routers
4. Authentication & Permissions
5. Browsable API

➢ How DRF Works

1. **Model** – represents database data
2. **Serializer** – converts data to JSON
3. **View / ViewSet** – handles logic (GET, POST, PUT, DELETE)
4. **URL/Router** – connects API endpoint
5. **Response** – sends JSON data to client

JSON

- JSON stands for JavaScript Object Notation.
- JSON is a way to send and receive data in a format that both humans and computers can understand.